Gemini Enterprise for Developers (2 Days)

Duration: 2 days

Audience: Software Engineers, Data Scientists, and Technical Leads who use Google Workspace, Gemini, and development environments in their daily work.

Track: Practical generative artificial intelligence for Developers.

Prerequisites: A Google account with access to Gemini and Google Workspace; basic programming knowledge; a sample script or codebase; access to the Antigravity IDE for Day 2.

Course Overview

Participants progress from connected data retrieval to structured prompting, technical document analysis, multimodal architecture work, and reusable code review assistants. The course then expands into interactive coding in Gemini Canvas, system architecture brainstorming, and concludes with a deep dive into Gemini Notebook for comprehensive codebase research and Antigravity IDE for autonomous agentic coding.

Main Topics Covered

- Connected Workspace retrieval for technical specs and API docs
- Cognitive frameworks for architectural decision support
- Deep synthesis of technical whitepapers and export of findings
- Multimodal analysis of architecture diagrams and UI components
- Persistent personas with Gems for code review and rubber duck debugging
- Interactive coding, refactoring, and inline debugging in Gemini Canvas
- Code Review & Security Auditing Workflows
- Architecture & API Brainstorming
- Gemini Notebook workspace setup for multi-file codebase ingestion
- Cross-document synthesis and technical documentation generation
- Agentic Coding with Antigravity IDE
- End-to-end Capstone developer project integration

Learning Outcomes

By the end of the course, participants will be able to:

- Retrieve and cross-reference PRDs and API specs from Gmail and Google Drive.
- Use a multi-persona prompt to evaluate tech stacks and architectural trade-offs.

- Extract, cite, and organize constraints from dense technical whitepapers.
- Analyze ER diagrams and user interfaces to generate foundational code blocks or identify friction.
- Create a reusable Code Reviewer Gem and perform complex prompt-based debugging.
- Use Gemini Canvas to refactor code, add logs, and fix bugs iteratively.
- Perform comprehensive security and logic reviews on pull requests.
- Build and interrogate a centralized research workspace containing legacy code and documentation using Gemini Notebook.
- Generate interactive API guides and technical documentation from Notebook sources.
- Transition from chat-based AI to agentic AI coding to automate multi-file tasks using Antigravity IDE.

Project Context / Tools

The course uses the Gemini web application, Gmail, Google Docs, Google Drive, Google Sheets, Gemini Gems, Gemini Canvas, Gemini Notebook, Antigravity IDE, and PDF/code file uploads. Participants should use non-sensitive material or prepared sample code repositories during the labs. Every lab is strictly modular; participants can drop in for any session and use their own ad-hoc context without needing the output of prior sessions.

Day 1: From Connected Data to Interactive Coding

1. Connected Workspace and Technical Data Retrieval

Concept & Demo: Move from manual searching to asking Gemini to retrieve and compare relevant technical information using extensions like `@Gmail` and `@Google Drive` to find PRDs and API specs. **Breakout Lab 1.1: The Spec Interrogation:** Participants search for a real project PRD or prepared sample with `@Gmail`, cross-reference it with a specific API document using `@Google Drive`, and produce a concise technical requirements report.

2. Cognitive Frameworks for Architecture

Concept & Demo: Use role assignments and competing viewpoints (e.g., security auditor vs. scalability expert) to debate architectural design decisions like "GraphQL vs REST". **Breakout Lab 2.1: The Architecture Board:** Participants select a technical decision, run a multi-persona prompt, check factual claims, and output an Architectural Decision Record (ADR) table outlining risks, rewards, and a recommendation.

3. Agentic Workflows and Technical Synthesis

Concept & Demo: Extract technical constraints or deprecation warnings from a dense API whitepaper, requiring citations and page numbers without summarizing. **Breakout Lab 3.1: The Docs Miner:** Participants upload a large technical PDF or text dump, run an extraction prompt, and export to Sheets to create a working table of technical constraints.

4. Multimodal Analysis for Developers

Concept & Demo: Work with sources beyond plain text. Analyze an Entity-Relationship (ER) diagram for single points of failure, or generate React/Tailwind code from a UI mockup screenshot. **Breakout Lab 4.1: The X-Ray Vision Test:** Participants upload an ER diagram, architecture map, or UI screenshot. They output either a vulnerability assessment or a scaffolded code block.

5. Persistent Personas and Code Reviewers

Concept & Demo: Introduce Gems as reusable assistants with saved instructions (e.g., "Strict Security Reviewer"). The Gem exposes logical gaps instead of just rewriting code. **Breakout Lab 5.1: Building the Pair Programmer:** Participants create a custom Gem for code review or unit test generation and test it on an independent code snippet.

6. Interactive Coding in Gemini Canvas

Concept & Demo: Use Gemini Canvas to "Add Comments", "Add Logs", and "Fix Bugs" iteratively on a provided script. **Breakout Lab 6.1: The Canvas Coder:** Participants bring a messy script into Gemini Canvas. They document the code, add robust logging, and interactively rewrite a core function.

Day 1 Session Breakdown Table

TOPIC	DURATION
1. Connected Workspace and Technical Data Retrieval (Concept & Demo)	40 min
Lab 1.1: The Spec Interrogation	20 min
2. Cognitive Frameworks for Architecture (Concept & Demo)	40 min
Lab 2.1: The Architecture Board	20 min
Morning Break	15 min
Kahoot Quiz 1	15 min
3. Agentic Workflows and Technical Synthesis (Concept & Demo)	40 min
Lab 3.1: The Docs Miner	20 min
Lunch	60 min
4. Multimodal Analysis for Developers (Concept & Demo)	40 min
Lab 4.1: The X-Ray Vision Test	20 min
Afternoon Break	15 min
Kahoot Quiz 2	15 min
5. Persistent Personas and Code Reviewers (Concept & Demo)	40 min
Lab 5.1: Building the Pair Programmer	20 min

TOPIC	DURATION
6. Interactive Coding in Gemini Canvas (Concept & Demo)	40 min
Lab 6.1: The Canvas Coder	20 min
Total	**8 hours**

Day 2: Advanced Workflows, Notebook, and Agentic Coding

7. Code Review & Debugging Workflows

Concept & Demo: Use Gemini to handle tedious development tasks by pasting a large Git diff or stack trace to identify edge cases or root causes. **Breakout Lab 7.1: The PR Auditor:** Participants supply a standalone pull request diff or stack trace. They prompt Gemini to summarize changes and identify one potential edge case or security flaw.

8. Architecture & API Brainstorming

Concept & Demo: Build comprehensive integration plans, such as designing a new RESTful API or planning a database schema migration. **Breakout Lab 8.1: The Integration Architect:** Participants outline a hypothetical feature. They generate an API contract (e.g., OpenAPI spec) and a step-by-step rollout plan.

9. Gemini Notebook: Foundation and Codebase Ingestion

Concept & Demo: Introduce Gemini Notebook as a centralized research workspace to minimize hallucinations. Ingest a small codebase along with API documentation PDFs. **Breakout Lab 9.1: The Workspace Builder:** Participants create a Notebook project and upload diverse technical materials (legacy module files, docs) to establish a codebase context.

10. Gemini Notebook: Synthesis and Active Interrogation

Concept & Demo: Ask cross-file questions and generate onboarding documentation or automated READMEs based purely on the uploaded source code. **Breakout Lab 10.1: The Documentation Generator:** Participants interrogate a loaded codebase, saving analytical notes and generating a comprehensive Developer Onboarding guide.

11. Agentic Coding with Antigravity IDE

Concept & Demo: Transition to Agentic AI. Give an AI agent a high-level goal in Antigravity IDE and watch it navigate the workspace, run tests, and write code autonomously. **Breakout Lab 11.1: The Autonomous Agent:** Participants set up a workspace in Antigravity IDE, issue a multi-file refactoring or test-generation command, and review the resulting diff.

12. Capstone Project & Workflow Integration

Concept & Demo: Integrate Canvas, Notebook, and Antigravity IDE into a daily engineering workflow.
Breakout Lab 12.1: End-to-End Application: Participants independently select a fresh development task and use their preferred combination of tools to produce a working feature, fix a bug, or document a system.

Day 2 Session Breakdown Table

TOPIC	DURATION
7. Code Review & Debugging Workflows (Concept & Demo)	40 min
Lab 7.1: The PR Auditor	20 min
8. Architecture & API Brainstorming (Concept & Demo)	40 min
Lab 8.1: The Integration Architect	20 min
Morning Break	15 min
Kahoot Quiz 3	15 min
9. Gemini Notebook: Foundation and Codebase Ingestion (Concept & Demo)	40 min
Lab 9.1: The Workspace Builder	20 min
Lunch	60 min
10. Gemini Notebook: Synthesis and Active Interrogation (Concept & Demo)	40 min
Lab 10.1: The Documentation Generator	20 min
Afternoon Break	15 min
Kahoot Quiz 4	15 min
11. Agentic Coding with Antigravity IDE (Concept & Demo)	40 min
Lab 11.1: The Autonomous Agent	20 min
12. Capstone Project & Workflow Integration (Concept & Demo)	40 min
Lab 12.1: End-to-End Application	20 min
Total	**8 hours**

Teaching Philosophy

Demonstrations prioritize making prompt structures visible and contrasting chat-based AI (Gemini Canvas) with agentic AI (Antigravity IDE). Every lab is strictly modular, allowing participants to use their own ad-hoc context without needing the output of prior sessions. Each lab ends with a usable technical artifact.